



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

UNIVERSITI TEKNOLOGI MALAYSIA  
FACULTY OF COMPUTING, UTM  
SEMESTER I, SESION 2023/2024

**ASSIGNMENT 1**

**SECI1113 : COMPUTATIONAL MATHEMATICS**

**SECTION 01**

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**COURSE : BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)**

**SUBMISSION DATE : 16 APRIL 2024**

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## SEC11113 : Computational Mathematics

Assignment 1

1. a)

$$\text{Matrix } A = \begin{bmatrix} 13 & 8 & 5 \\ 15 & 12 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

$$b) A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

$$\begin{aligned} |A| &= 13 \begin{vmatrix} 12 & 10 \\ 17 & 15 \end{vmatrix} - 8 \begin{vmatrix} 15 & 10 \\ 20 & 15 \end{vmatrix} + 5 \begin{vmatrix} 15 & 12 \\ 20 & 17 \end{vmatrix} \\ &= 13(10) - 8(25) + 5(15) \\ &= 130 - 200 + 75 \\ &= 5 \end{aligned}$$

$$\text{cofactor matrix : } \begin{bmatrix} \begin{vmatrix} 12 & 10 \\ 17 & 15 \end{vmatrix} & - \begin{vmatrix} 15 & 10 \\ 20 & 15 \end{vmatrix} & \begin{vmatrix} 15 & 12 \\ 20 & 17 \end{vmatrix} \\ - \begin{vmatrix} 8 & 5 \\ 17 & 15 \end{vmatrix} & \begin{vmatrix} 13 & 5 \\ 20 & 15 \end{vmatrix} & - \begin{vmatrix} 13 & 8 \\ 20 & 17 \end{vmatrix} \\ \begin{vmatrix} 8 & 5 \\ 13 & 10 \end{vmatrix} & - \begin{vmatrix} 13 & 5 \\ 15 & 10 \end{vmatrix} & \begin{vmatrix} 13 & 8 \\ 15 & 12 \end{vmatrix} \end{bmatrix}$$

$$\text{Adjoint } A : \begin{bmatrix} 10 & -25 & 15 \\ -35 & 95 & -61 \\ 20 & -55 & 36 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 10 & -35 & 20 \\ -25 & 95 & -55 \\ 15 & -61 & 36 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -7 & 4 \\ -5 & 19 & -11 \\ 3 & -\frac{61}{5} & \frac{36}{5} \end{bmatrix}$$



$$2. \begin{pmatrix} 2 & 3 & 1 \\ 3 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} p \\ m \\ b \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 10 \end{pmatrix}$$

$$\begin{array}{ccc|c} 2 & 3 & 1 & 5 \\ 3 & 4 & 2 & 8 \\ 1 & 2 & 3 & 10 \end{array} \xrightarrow{R_3 \leftrightarrow R_1} \begin{array}{ccc|c} 1 & 2 & 3 & 10 \\ 3 & 4 & 2 & 8 \\ 2 & 3 & 1 & 5 \end{array}$$

$$m_{21} = -\frac{3}{1} \\ = -3$$

$$R_2 \rightarrow R_2 - 3R_1 \downarrow$$

$$\begin{array}{ccc|c} 1 & 2 & 3 & 10 \\ 0 & -2 & -7 & -22 \\ 2 & 3 & 1 & 5 \end{array}$$

$$m_{31} = -\frac{2}{1} \\ = -2$$

$$R_3 \rightarrow R_3 - 2R_1 \downarrow$$

$$\begin{array}{ccc|c} 1 & 2 & 3 & 10 \\ 0 & -2 & -7 & -22 \\ 0 & -1 & -5 & -15 \end{array}$$

$$R_3 \rightarrow 1 \cdot R_3$$

$$R_3 \leftrightarrow R_2 \downarrow$$

$$\begin{array}{ccc|c} 1 & 2 & 3 & 10 \\ 0 & 1 & 5 & 15 \\ 0 & -2 & -7 & -22 \end{array}$$

$$m_{32} = -\frac{-2}{1} \\ = 2$$

$$R_3 \rightarrow R_3 + 2(R_2) \downarrow$$

$$\begin{array}{ccc|c} 1 & 2 & 3 & 10 \\ 0 & 1 & 5 & 15 \\ 0 & 0 & 3 & 8 \end{array}$$

$$p + 2m + 3b = 10 \quad \text{--- ①}$$

$$m + 5b = 15 \quad \text{--- ②}$$

$$3b = 8 \quad \text{--- ③}$$

$$b = \frac{8}{3}$$

$$m + 5\left(\frac{8}{3}\right) = 15$$

$$m + \frac{40}{3} = 15$$

$$m = \frac{5}{3}$$

$$p + 2\left(\frac{5}{3}\right) + 3\left(\frac{8}{3}\right) = 10$$

$$p + \frac{34}{3} = 10$$

$$p = -\frac{4}{3}$$

$$\therefore p = -\frac{4}{3}$$

$$m = \frac{5}{3}$$

$$b = \frac{8}{3}$$

3.  $P = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & -1 \\ 3 & 6 & 0 \end{bmatrix}$   $|P| = 3 \begin{vmatrix} 2 & -1 \\ 4 & -1 \end{vmatrix} - 6 \begin{vmatrix} 1 & -1 \\ 3 & 0 \end{vmatrix} + 0 \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$   
 $= 3(-6) - 6(-3)$   
 $= -18 + 18$   
 $= 0$

$\therefore$  Matrices  $P$  does not have an inverse

$Q = \begin{bmatrix} 2 & 7 & 4 \\ 5 & 19 & 11 \\ 3 & 13 & 8 \end{bmatrix}$   $|Q| = 2 \begin{vmatrix} 19 & 11 \\ 13 & 8 \end{vmatrix} - 7 \begin{vmatrix} 5 & 11 \\ 3 & 8 \end{vmatrix} + 4 \begin{vmatrix} 5 & 19 \\ 3 & 13 \end{vmatrix}$   
 $= 2(9) - 7(7) + 4(8)$   
 $= 1$

Matrices  $Q$  have an inverse

$Q^{-1} =$

$$\left[ \begin{array}{ccc|ccc} 2 & 7 & 4 & 1 & 0 & 0 \\ 5 & 19 & 11 & 0 & 1 & 0 \\ 3 & 13 & 8 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left[ \begin{array}{ccc|ccc} 1 & \frac{7}{2} & 2 & \frac{1}{2} & 0 & 0 \\ 5 & 19 & 11 & 0 & 1 & 0 \\ 3 & 13 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{R_2 \rightarrow R_2 - 5R_1} \downarrow \left[ \begin{array}{ccc|ccc} 1 & \frac{7}{2} & 2 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{3}{2} & 1 & -\frac{5}{2} & 1 & 0 \\ 0 & \frac{5}{2} & 2 & -\frac{3}{2} & 0 & 1 \end{array} \right] \xleftarrow{R_3 \rightarrow R_3 - 3R_1} \left[ \begin{array}{ccc|ccc} 1 & \frac{7}{2} & 2 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{3}{2} & 1 & -\frac{5}{2} & 1 & 0 \\ 0 & 13 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$\downarrow \xrightarrow{R_2 \rightarrow \frac{2}{3}R_2} \left[ \begin{array}{ccc|ccc} 1 & \frac{7}{2} & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{5}{3} & \frac{2}{3} & 0 \\ 0 & \frac{5}{2} & 2 & -\frac{3}{2} & 0 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - \frac{5}{2}R_2} \left[ \begin{array}{ccc|ccc} 1 & \frac{7}{2} & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{5}{3} & \frac{2}{3} & 0 \\ 0 & 0 & \frac{1}{3} & \frac{8}{3} & -\frac{5}{3} & 1 \end{array} \right]$$

$$\downarrow \xrightarrow{R_3 \rightarrow 3R_3} \left[ \begin{array}{ccc|ccc} 1 & 0 & -\frac{1}{3} & \frac{19}{3} & -\frac{7}{3} & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{5}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & 8 & -5 & 3 \end{array} \right] \xleftarrow{R_1 \rightarrow R_1 - \frac{7}{2}R_2} \left[ \begin{array}{ccc|ccc} 1 & \frac{7}{2} & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{2}{3} & -\frac{5}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & 8 & -5 & 3 \end{array} \right]$$

$$\downarrow \xrightarrow{R_1 \rightarrow R_1 + \frac{1}{3}R_3} \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 9 & -4 & 1 \\ 0 & 1 & \frac{2}{3} & -\frac{5}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & 8 & -5 & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - \frac{2}{3}R_3} \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 9 & -4 & 1 \\ 0 & 1 & 0 & -7 & 4 & -2 \\ 0 & 0 & 1 & 8 & -5 & 3 \end{array} \right]$$

$$Q^{-1} = \begin{pmatrix} 9 & -4 & 1 \\ -7 & 4 & -2 \\ 8 & -5 & 3 \end{pmatrix}$$



4.

$$A = \begin{bmatrix} 4 & 2 & 3 \\ 3 & 3 & 2 \\ 2 & 4 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 18 \\ 16 \\ 17 \end{bmatrix}$$

(i) Doolittle

$$\begin{matrix} A \\ \begin{bmatrix} 4 & 2 & 3 \\ 3 & 3 & 2 \\ 2 & 4 & 2 \end{bmatrix} \end{matrix} = \begin{matrix} \begin{bmatrix} d & e & f \\ ad & ae+g & af+h \\ bd & be+cg & bf+ch+i \end{bmatrix} \\ \begin{bmatrix} d=4 & e=2 & f=3 \\ a=\frac{3}{4} & g=\frac{3}{2} & h=-\frac{1}{4} \\ b=\frac{1}{2} & c=2 & i=1 \end{bmatrix} \end{matrix}$$

$$LY = B$$

$$\begin{pmatrix} 1 & 0 & 0 \\ \frac{3}{4} & 1 & 0 \\ \frac{1}{2} & 2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 18 \\ 16 \\ 17 \end{pmatrix}$$

$$y_1 = 18$$

$$\frac{3}{4}y_1 + y_2 = 16$$

$$\frac{3}{4}(18) + y_2 = 16$$

$$y_2 = \frac{5}{2}$$

$$\frac{1}{2}y_1 + 2y_2 + y_3 = 17$$

$$\frac{1}{2}(18) + 2\left(\frac{5}{2}\right) + y_3 = 17$$

$$y_3 = 3$$

$$UX = Y$$

$$\begin{pmatrix} 4 & 2 & 3 \\ 0 & \frac{3}{2} & -\frac{1}{4} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 18 \\ \frac{5}{2} \\ 3 \end{pmatrix}$$

$$x_3 = 3$$

$$\frac{3}{2}x_2 - \frac{1}{4}x_3 = \frac{5}{2}$$

$$\frac{3}{2}x_2 - \frac{1}{4}(3) = \frac{5}{2}$$

$$x_2 = \frac{13}{6}$$

$$4x_1 + 2x_2 + 3x_3 = 18$$

$$4x_1 + 2\left(\frac{13}{6}\right) + 3(3) = 18$$

$$4x_1 = \frac{14}{3}$$

$$x_1 = \frac{7}{6}$$

$$\therefore x = \begin{bmatrix} \frac{7}{6} & \frac{13}{6} & 3 \end{bmatrix}^T$$

4. (ii) Crout method

$$\begin{aligned}
 & \begin{matrix} & A \\ \begin{pmatrix} 4 & 2 & 3 \\ 3 & 3 & 2 \\ 2 & 4 & 2 \end{pmatrix} & = \begin{pmatrix} a & ag & ah \\ b & bg+c & bh+ci \\ d & dg+e & dh+ei+f \end{pmatrix} \\ & = \begin{pmatrix} a=4 & g=\frac{1}{2} & h=\frac{3}{4} \\ b=3 & c=\frac{3}{2} & i=-\frac{1}{6} \\ d=2 & e=3 & f=1 \end{pmatrix}
 \end{aligned}$$

$$LY = B$$

$$\begin{pmatrix} 4 & 0 & 0 \\ 3 & \frac{3}{2} & 0 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 18 \\ 16 \\ 17 \end{pmatrix}$$

$$4y_1 = 18$$

$$y_1 = \frac{9}{2}$$

$$3y_1 + \frac{3}{2}y_2 = 16$$

$$3\left(\frac{9}{2}\right) + \frac{3}{2}y_2 = 16$$

$$y_2 = \frac{5}{3}$$

$$2y_1 + 3y_2 + y_3 = 17$$

$$2\left(\frac{9}{2}\right) + 3\left(\frac{5}{3}\right) + y_3 = 17$$

$$y_3 = 3$$

$$UX = Y$$

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{4} \\ 0 & 1 & -\frac{1}{6} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \frac{9}{2} \\ \frac{5}{3} \\ 3 \end{pmatrix}$$

$$x_3 = 3$$

$$x_2 - \frac{1}{6}x_3 = \frac{5}{3}$$

$$x_2 - \frac{1}{6}(3) = \frac{5}{3}$$

$$x_2 = \frac{13}{6}$$

$$x_1 + \frac{1}{2}x_2 + \frac{3}{4}x_3 = \frac{9}{2}$$

$$x_1 + \frac{1}{2}\left(\frac{13}{6}\right) + \frac{3}{4}(3) = \frac{9}{2}$$

$$x_1 = \frac{7}{6}$$

$$\therefore x = \left[ \frac{7}{6} \quad \frac{13}{6} \quad 3 \right]^T$$



5.

$$u = (5, 4, -3)$$

$$v = (1, -2, -1)$$

a) Euclidean distance

$$\begin{aligned} d(u, v) &= \sqrt{(5-1)^2 + (4+2)^2 + (-3+1)^2} \\ &= \sqrt{56} \\ &= 2\sqrt{14} \end{aligned}$$

b) Dot product,  $\theta = 45^\circ$ 

$$u \cdot v = |u| |v| \cos 45^\circ$$

$$\begin{aligned} |u| &= \sqrt{5^2 + 4^2 + (-3)^2} \\ &= \sqrt{50} \\ &= 5\sqrt{2} \end{aligned}$$

$$\begin{aligned} |v| &= \sqrt{1^2 + (-2)^2 + (-1)^2} \\ &= \sqrt{6} \end{aligned}$$

$$\begin{aligned} u \cdot v &= (5\sqrt{2})(\sqrt{6}) \cos 45^\circ \\ &= 10\sqrt{3} \cos 45^\circ \\ &= 5\sqrt{6} \end{aligned}$$

c) Cross product

$$u \times v = \begin{vmatrix} i & j & k \\ 5 & 4 & -3 \\ 1 & -2 & -1 \end{vmatrix}$$

$$\begin{aligned} u \times v &= i \begin{vmatrix} -3 & -1 \\ -2 & -1 \end{vmatrix} - j \begin{vmatrix} 5 & -3 \\ 1 & -1 \end{vmatrix} + k \begin{vmatrix} 5 & 4 \\ 1 & -2 \end{vmatrix} \\ &= -10i + 2j - 14k \end{aligned}$$

6.  $v_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$  ;  $v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$  ;  $v_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$  and  $v_4 = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$

linear combination

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$$

$$\tilde{A} = \left[ \begin{array}{ccc|c} 1 & 2 & 0 & -2 \\ -1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 \end{array} \right] \xrightarrow{\begin{array}{l} R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}} \left[ \begin{array}{ccc|c} 1 & 2 & 0 & -2 \\ 0 & 3 & 2 & 0 \\ 0 & -1 & 1 & 3 \end{array} \right]$$

$$\xrightarrow{R_2 \rightarrow \frac{1}{3}R_2} \left[ \begin{array}{ccc|c} 1 & 2 & 0 & -2 \\ 0 & 1 & \frac{2}{3} & 0 \\ 0 & -1 & 1 & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_2} \left[ \begin{array}{ccc|c} 1 & 2 & 0 & -2 \\ 0 & 1 & \frac{2}{3} & 0 \\ 0 & 0 & \frac{5}{3} & 3 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow \frac{3}{5}R_3} \left[ \begin{array}{ccc|c} 1 & 2 & 0 & -2 \\ 0 & 1 & \frac{2}{3} & 0 \\ 0 & 0 & 1 & \frac{9}{5} \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[ \begin{array}{ccc|c} 1 & 0 & -\frac{4}{3} & -2 \\ 0 & 1 & \frac{2}{3} & 0 \\ 0 & 0 & 1 & \frac{9}{5} \end{array} \right]$$

$$\xrightarrow{R_1 \rightarrow R_1 + \frac{4}{3}R_3} \left[ \begin{array}{ccc|c} 1 & 0 & 0 & \frac{2}{5} \\ 0 & 1 & \frac{2}{3} & 0 \\ 0 & 0 & 1 & \frac{9}{5} \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - \frac{2}{3}R_3} \left[ \begin{array}{ccc|c} 1 & 0 & 0 & \frac{2}{5} \\ 0 & 1 & 0 & -\frac{6}{5} \\ 0 & 0 & 1 & \frac{9}{5} \end{array} \right]$$

$$\begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$$

$$= \frac{2}{5} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} - \frac{6}{5} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \frac{9}{5} \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$$



7.  $u = (1, -3, 2)$   $v = (2, -1, 1)$

a)

$$\begin{pmatrix} 1 \\ k \\ 5 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$$\left( \begin{array}{cc|c} 1 & 2 & 1 \\ -3 & -1 & k \\ 2 & 1 & 5 \end{array} \right) \xrightarrow[\begin{array}{l} R_2 \rightarrow R_2 + 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}]{\hspace{1cm}} \left( \begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 5 & k+3 \\ 0 & -3 & 3 \end{array} \right)$$

$$-3c_2 = 3$$

$$c_2 = -1$$

$$c_1 + 2c_2 = 1$$

$$c_1 + 2(-1) = 1$$

$$c_1 = 3$$

$$5c_2 = k+3$$

$$5(-1) = k+3$$

$$k = -8$$

$$\therefore k = -8$$

b)  $\begin{pmatrix} 1 & 2 \\ -3 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -8 \\ 5 \end{pmatrix}$

$$c_1 = 3 (\neq 0)$$

$$c_2 = -1 (\neq 0)$$

$\therefore u$  and  $v$  are not linearly independent